A Physics-Guided Hybrid System for 5G Root-Cause Diagnosis: Champion of the AI Telco Troubleshooting Challenge

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***Abstract*—We report the champion solution of *The AI Telco Troubleshooting Challenge* (Zindi, 2026), a global competition (200+ teams) on 5G root-cause classification. Frontier LLMs plateau below 77% accuracy on this task; our 17-rule, physics-guided decision engine reaches 97.1% on standard telecom questions (93.85% overall), invoking an LLM on only ∼27% of samples. We show that (i) LLMs are misaligned with the numerical operators this task requires (cross-row min, count, sum), and no prompting strategy closes the gap; (ii) RAG context cannot lift a model above its reasoning ceiling; (iii) fine-tuning improves domain accuracy but triggers catastrophic forgetting on the non-telecom questions the benchmark deliberately includes; and (iv) a *narrow-then-reason* hybrid—rules eliminate impossible options, an un-fine-tuned LLM resolves the residual ambiguity—systematically dominates either component alone. The track championship (reproducibility-verified) provides strong external validation.**

***Index Terms*—5G, root-cause analysis, large language models, neuro-symbolic AI, rule-based systems, hybrid AI, network operations.**

I. Introduction

The dominant narrative of 2024–2026 holds that a sufficiently large language model, properly prompted or retrieval-augmented, will subsume narrow expert systems. *The AI Telco Troubleshooting Challenge* [1]—a global Zindi competition on 5G fault diagnosis—offers a clean falsification of this view in one practically important domain.

In this paper we describe the methodology that won Track 1 of that competition. The system is, by 2026 standards, almost embarrassingly small: a 17-rule decision engine over a nine-dimensional feature vector, with an un-fine-tuned Qwen3-32B invoked only on the residual ambiguous cases. Yet it outperforms every frontier LLM we benchmarked (Claude Opus 4.5, Gemini 3 Pro, DeepSeek) by 20+ percentage points, achieving the track championship on the held-out Phase-2 test set.

Contributions.

1) A reproducible 9-feature engineering pipeline for 5G drive-test and engineering-parameter tables, including a domain-specific data-cleaning rule (Digital Tilt code $255 \to 6°$) that no LLM identified unaided.
2) A priority-ordered rule engine with 17 rules grounded in 3GPP physics; six of eight root causes are resolved by deterministic single-feature rules, leaving only the C1/C3 boundary for multi-feature logic.
3) A *narrow-then-reason* hybrid architecture in which rules eliminate impossible candidates and an LLM resolves only the residual ambiguity, reducing API calls by 73%.
4) Empirical evidence that (a) frontier LLMs hit a hard ceiling below 77% on this task, and (b) smaller LLMs cannot close the gap by adding RAG context—a capability ceiling that has direct consequences for system design.
5) External validation via track championship (MWC 2026, reproducibility-verified).

II. Related Work

A. Telecom Fault Diagnosis

Automated root-cause analysis (RCA) for mobile networks has a long pre-LLM history. Barco et al. [5] applied Bayesian networks to GSM/UMTS alarm correlation, demonstrating that probabilistic graphical models can localise faults when the causal structure is known. Zoha et al. [6] surveyed data-driven self-healing in LTE, covering anomaly detection via random forests, gradient boosting, and clustering over KPI time-series. Mulvey et al. [7] extended these methods to 5G cell outage detection. A common thread is that classical ML works well on structured, low-dimensional KPI features—a property our rule engine exploits directly.

B. LLMs for Structured and Numerical Reasoning

Recent work has systematically exposed LLM limitations on tasks requiring precise numerical or tabular reasoning. Sui et al. [8] showed that even GPT-4-class models struggle with structured table understanding when column alignment is disrupted. Chen et al. [9] demonstrated that LLMs cannot reliably self-correct reasoning errors without external feedback. Shi et al. [10] found that LLM accuracy degrades significantly in the presence of irrelevant context—a direct analogue of the long pipe-delimited tables in our task. Program-aided approaches [11], [12] circumvent arithmetic limitations by delegating computation to code interpreters, but they require structured API access and add latency —constraints that our lightweight rule engine avoids entirely.

C. Neuro-Symbolic Hybrid Systems

The broader neuro-symbolic AI programme seeks principled integration of neural learning with symbolic reasoning. Garcez and Lamb [13] provide a foundational survey; Marcus [14] argues that hybrid architectures are necessary for robust AI. Mao et al. [15] combine neural perception with symbolic program execution on visual QA. Our work instantiates this philosophy in a production telecom setting: the rule engine provides the symbolic backbone, while the LLM contributes flexible reasoning on residual ambiguity. Unlike end-to-end differentiable neuro-symbolic systems, our architecture is fully modular—the rule engine operates independently and the LLM is interchangeable.

III. Problem Setting and Competition Context

A. Task Definition

The Phase-2 test set contains 863 questions drawn from two domains. Approximately 80% are *standard telecom* questions: each consists of (a) a drive-test (telelog) table with time-stamped KPIs (SS-RSRP, SINR, serving/neighbor PCI, GPS speed, DL RB count) and (b) an engineering-parameter table (mechanical/digital tilt, PCI), and the model must classify the throughput-degradation root cause into one of eight classes C1–C8 (detailed in Table II). The remaining $\sim$20% are *non-standard telecom* questions (different table schemas) or *general-knowledge* questions unrelated to 5G. This mixed-domain design is a critical feature of the benchmark: it penalises fine-tuning approaches that sacrifice generalist capability for domain-specific gains (Section VIII-C).

B. Competition as a Methodology Benchmark

We emphasise the competition design because it constitutes an unusually strong external benchmark of *methodologies*, not merely models. Monetary incentives, a public leaderboard, and a deliberate Phase-1/Phase-2 distribution shift penalise overfitting. Critically, the final ranking was determined not by leaderboard score alone but by a *reproducibility review*: teams whose solutions could not be independently verified on unseen data were disqualified. This design inherently penalises both overfitting and narrow fine-tuning (Section VIII-C).

IV. Why Frontier LLMs Hit a Ceiling

A. Operator Mismatch

The task masquerades as language understanding but its atomic operators are numerical: *count* PCI changes, *argmin* of an RSRP column, *sum* of tilt fields. LLMs tokenize pipe-delimited numeric tables in ways that break column alignment, and their cross-row arithmetic is empirically unreliable beyond short sequences.

B. The Frontier Ceiling

Our first attempt used an un-fine-tuned Qwen3-32B with a KNN-based RAG pipeline retrieving similar solved cases from the training set. This configuration reached $\sim$60% accuracy—a reasonable baseline, but far from deployable. Switching to frontier-class models (Claude Opus 4.5, Gemini 3 Pro) with extensive prompt engineering, chain-of-thought, and tool-use scaffolding pushed accuracy up to $\sim$77%, where it saturated regardless of further prompt refinement. The 17-percentage-point gap between the 32B baseline and the frontier ceiling is attributable to reasoning capacity, not context quality. Our hybrid approach avoids this trade-off entirely.

TABLE I
Nine-Dimensional Diagnostic Feature Vector

Feature	Extraction Logic	Physical Meaning
$r_{\min}$	Min Serving SS-RSRP	Worst signal strength
$\theta_{\max}$	Max (Mech.+Dig. Tilt)	Most aggressive downtilt
$\theta_{\min}$	Min (Mech.+Dig. Tilt)	Least aggressive downtilt
θ_{Σ}	Sum of unique tilt values	Overall tilt budget
h	# Serving PCI changes	Handover frequency
n	Distinct Top-1 Nbr PCIs	Strong interfering cells
$v_{\max}$	Max GPS Speed	UE mobility level
$\overline{rb}$	Mean DL RB Num	Resource utilization
π_{conf}	Any PCI mod 30 collision (Boolean)	DMRS collision risk

C. The Small-Model RAG Trap

A natural follow-up question is whether enriching the smaller model's context can close the gap. We tested this by supplying Qwen3-32B with the same heavy RAG context (more retrieved examples, structured summaries, explicit reasoning templates) that benefited the frontier models. Accuracy improved only to $\sim$70%—still well below the frontier ceiling. The lesson generalises: *capability ceilings are properties of the model, not of its prompt*. Wielding LLMs effectively begins with mapping each candidate model's ceiling on the target operators, then routing only ceiling-fit subtasks to it—not assuming RAG is a universal lift.

V. Methodology

A. Iterative Trajectory

Our solution evolved through three regimes (see also Table III). The pure-LLM regime began with Qwen3-32B + RAG at $\sim$60% accuracy on standard telecom questions, then improved to $\sim$77% with frontier models (Claude Opus 4.5, Gemini 3 Pro), where it saturated. The decisive leap came from injecting physics-grounded rules: the 17-rule hybrid system reached 97.1% on standard telecom questions (93.85% overall), breaking the LLM ceiling by 20+ percentage points. This trajectory illustrates the core thesis: LLM accuracy saturates on structured-numerical tasks, and escaping the plateau requires domain knowledge injected as rules, not larger models or richer prompts.

B. Feature Engineering

We extract a 9-dimensional diagnostic vector $\mathbf{x} = [r_{\min}, \theta_{\max}, \theta_{\min}, \theta_{\Sigma}, h, n, v_{\max}, \overline{rb}, \pi_{\text{conf}}]$. Table I lists each feature's extraction logic and physical meaning.

A non-obvious cleaning step proved critical: when the Digital Tilt field contains the raw value 255, its semantic meaning is 6°. This was identified by joint analysis of vendor configuration exports and training-set tilt distributions—no LLM flagged it unaided.

TABLE II
COMPLETE PRIORITY-ORDERED DECISION RULES (FIRST-MATCH SEMANTICS). RULES 1–6 ARE SINGLE-FEATURE SEPARATORS COVERING $\sim$62% OF SAMPLES AT $\sim$100% PRECISION. RULES 7–14 RESOLVE UNAMBIGUOUS C1/C3 REGIONS. RULES 15–17 FLAG RESIDUAL AMBIGUITY FOR LLM ADJUDICATION.

Pri.	Condition	Root Cause	Conf.
1	$n \geq 3$	C4: No dominant server	H
2	$h \geq 3$	C5: Frequent handovers	H
3	$h = 2$	C2: Overshooting	H
4	$v_{\max} > 40$ km/h	C7: High-speed mobility	H
5	$\overline{rb} < 170$	C8: RB starvation	H
6	π_{conf}	C6: PCI mod-30 conflict	H
7	$\theta_{\max} < 12°$	C3: Under-tilted	H
8	$\theta_{\Sigma} < 19°$	C3: Under-tilted	H
9	$\theta_{\min} < 6°$	C3: Under-tilted	H
10	$\theta_{\min} < 10 \wedge r_{\min} > -89$	C3: Under-tilted	H
11	$r_{\min} < -90$ dBm	C1: Weak coverage	H
12	$\theta_{\max} > 29°$	C1: Over-tilted	H
13	$\theta_{\Sigma} > 52°$	C1: Over-tilted	H
14	$\theta_{\min} > 25°$	C1: Over-tilted	H
15	$r_{\min} < -88.5 \wedge \theta_{\Sigma} > 39 \wedge \theta_{\max} \geq 22$	C1 (tentative)	L
16	$\theta_{\min} < 10 \vee \theta_{\Sigma} \leq 35$	C3 (tentative)	L
17	otherwise	C3 (default)	L

C. Discriminative Statistics

Class-conditional statistics on the Phase-1 ground truth revealed three exploitable structures: (i) $n \geq 3$ is a sufficient condition for C4 (overlapping coverage with no dominant server); (ii) although PCI mod-30 collisions appear across multiple classes, the priority ordering of rules 1–5 strips away all non-C6 cases that happen to exhibit collisions, making the residual collision flag a reliable C6 discriminator; (iii) C1 and C3 overlap in single-feature space and require multi-feature boundaries.

D. Priority-Ordered Rule Engine

The engine comprises 17 priority-ordered rules executed with *first-match semantics*: each sample is classified by the highest-priority rule whose condition it satisfies. The ordering reflects a deliberate design principle: *discrete countable events* (handover count, neighbor count) are checked first because they admit exact thresholds with near-zero ambiguity; *continuous threshold judgments* (tilt angles, RSRP levels) are deferred because their decision boundaries are inherently fuzzier and require multi-feature logic. Table II summarises all 17 decision rules.

Rules 1–6 (single-feature separators). Five root causes—C4 (no dominant server), C5 (frequent handover), C2 (overshooting), C7 (high speed), C8 (RB starvation)—each admit a single feature with near-perfect class separation. C6 (PCI mod-30 conflict) is resolved by a Boolean collision flag. Together these six rules cover $\sim$62% of all samples at $\sim$100% precision.

Rules 7–14 (C1/C3 unambiguous extremes). After rules 1–6, the remaining samples are predominantly C1 or C3—physically opposite ends of the antenna-tilt spectrum. Rules 7–10 capture the unambiguous C3 (under-tilted) region: low maximum tilt ($\theta_{\max} < 12°$), low total tilt ($\theta_{\Sigma} < 19°$), very low minimum tilt ($\theta_{\min} < 6°$), or low tilt combined with adequate signal ($\theta_{\min} < 10° \wedge r_{\min} > -89$ dBm). Rules 11–14 capture the unambiguous C1 (over-tilted/weak-coverage) region: low RSRP ($r_{\min} < -90$ dBm), extreme downtilt ($\theta_{\max} > 29°$), high total tilt ($\theta_{\Sigma} > 52°$), or uniformly high tilt ($\theta_{\min} > 25°$). Thresholds were calibrated via a four-step process: (i) pairwise scatter plots of C1 vs. C3 training samples (Fig. 1), (ii) physical reasonability checks against 3GPP conventions, (iii) iterative validation on held-out Phase-1 data, and (iv) domain-expert adjudication.

Rules 15–17 (residual ambiguity). Intermediate cases falling between the C1 and C3 extremes are assigned a tentative label with low confidence and forwarded to the LLM for adjudication (Section V-E).

1) C1 vs. C3: The Core Diagnostic Challenge: C1 (over-tilted serving cell, weak far-end coverage) and C3 (under-tilted serving cell, neighbour offers higher throughput) are physically *opposite ends* of the same parameter—antenna down-tilt—and consequently overlap heavily in any single feature dimension.

Figure 1 shows pairwise scatter plots of C1 (red) and C3 (blue) training samples across four tilt and RSRP features. Each feature separates the two classes at the extremes—e.g., $r_{\min} < -90$ dBm is almost purely C1, $\theta_{\max} < 12°$ almost purely C3—but a broad interior region contains both classes. The multi-feature rules 7–17 directly mirror these findings: each rule peels off an unambiguous region, progressively shrinking the residual overlap until only truly ambiguous cases remain for LLM adjudication.

E. Narrow-then-Reason Hybrid

The full pipeline (Fig. 2) embodies our second core claim: hybrid systems systematically dominate either pure component. The rule layer's role is not always to *decide*—often it is to *eliminate*, narrowing eight options to two or three before invoking the LLM with a constrained prompt. This both raises LLM accuracy on the residual cases and slashes API cost.

LLM component. The hybrid uses an un-fine-tuned Qwen3-32B as the LLM component. For standard telecom questions reaching rules 15–17, the prompt presents only the two surviving candidates (C1 and C3) together with the extracted feature vector, reducing the LLM's task from 8-way to binary classification. For non-standard telecom questions (different table schemas), a small set of RSRP/SINR threshold pre-filters is applied before invoking the LLM with a domain-specific prompt. General-knowledge questions are routed directly to the LLM without rule-based pre-processing. If the LLM call fails (timeout or rate-limiting), the system falls back to the rule engine's tentative label.

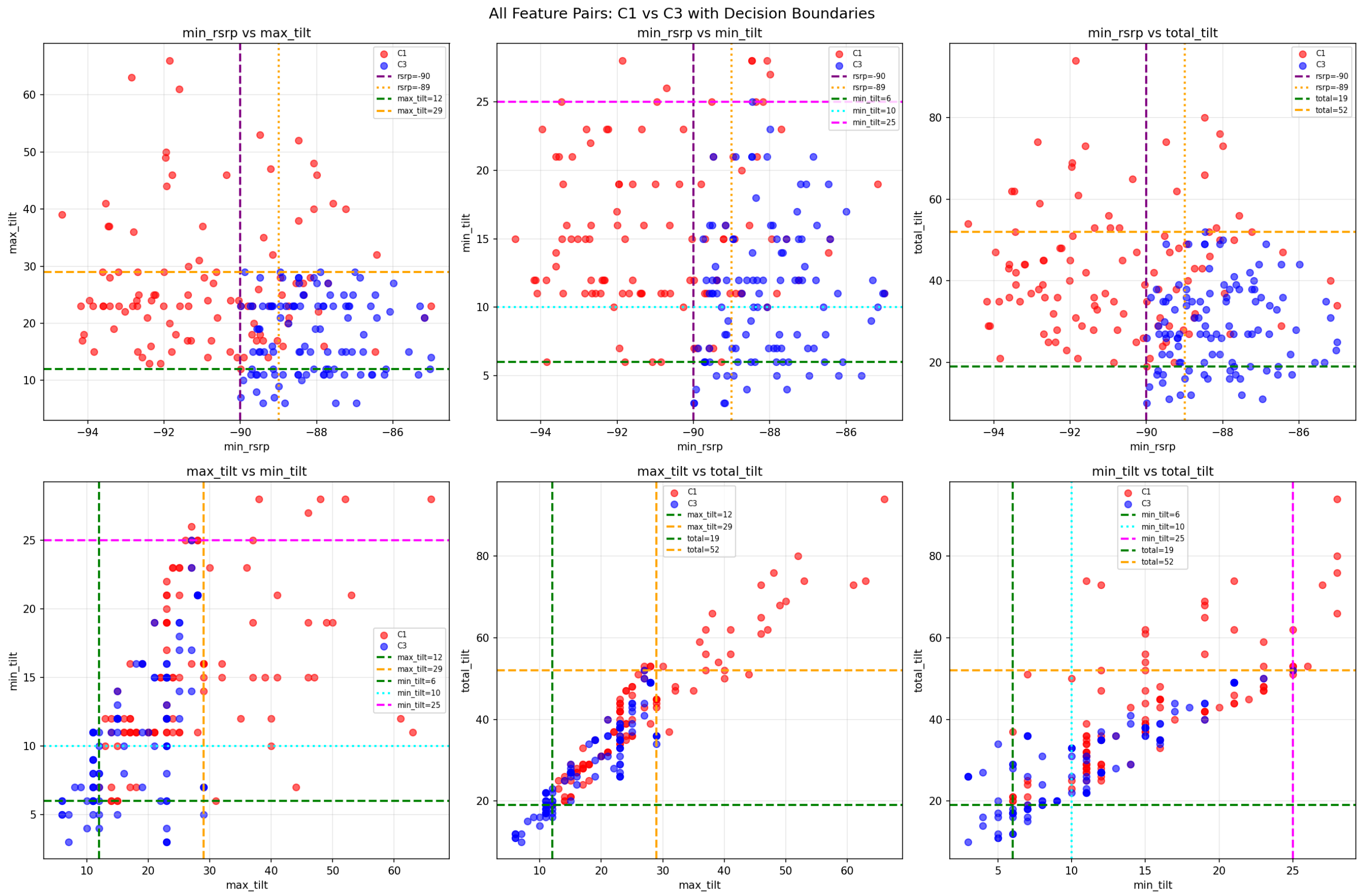

Fig. 1. Pairwise scatter plots of C1 (red) and C3 (blue) training samples across four tilt and RSRP features, with physics-grounded decision boundaries. Each dashed line corresponds to a rule threshold. The overlapping interior motivates the multi-feature, multi-tier design of the rule engine.

VI. Human–LLM Co-Design of the Rule Base

We disclose—and theorize—the role of LLMs in *building* the system. Rule candidates, confusion-pattern summaries, and option-pruning heuristics were drafted with Claude Opus 4.5 acting as a reasoning partner; humans supplied physical priors, anomaly detection (e.g., the 255-encoding), and final threshold adjudication. We name this pattern *LLM-assisted symbolic system synthesis*: the LLM is neither the predictor nor a black-box oracle, but a collaborator in producing a small, auditable, human-owned artefact.

VII. Results

Table III consolidates the comparison. Because the six deterministic rules (1–6) are provably correct given accurate feature extraction, the only classification uncertainty lies in the C1/C3 boundary; all other root causes are resolved without LLM involvement. Notably, the hybrid's LLM component is Qwen3-32B—the same model that achieves only ∼60% when used alone. The 34-percentage-point lift (60% → 93.85%) is entirely attributable to the rule engine and the narrow-then-reason architecture, not to model scale.

TABLE III
Method Comparison. *Telecom Acc.* is measured on standard-format telecom questions (∼80% of the test set); *Overall Acc.* includes non-standard telecom and general-knowledge questions. The competition ranking was determined on a held-out Phase-2 test set whose ground truth was never disclosed to participants.

Method	Telecom Acc.	Overall Acc.	LLM calls
Qwen3-32B + RAG	∼60%	—	100%
Qwen3-32B + heavy RAG	∼70%	—	100%
Gemini 3 Pro (best)	<77%	—	100%
Claude Opus 4.5 (best)	<77%	—	100%
Hybrid (ours)†	**97.1%**	**93.85%**	**27%**
Competition outcome (held-out Phase-2 test set):			
Hybrid (ours)	**Track 1 Champion (reprod.-verified)**		

†LLM component: un-fine-tuned Qwen3-32B.

VIII. Discussion

A. When Rule Engines Win

The conditions favouring our paradigm are: (i) low-dimensional structured inputs, (ii) strong physical priors with well-known thresholds, (iii) a closed and small label set, and

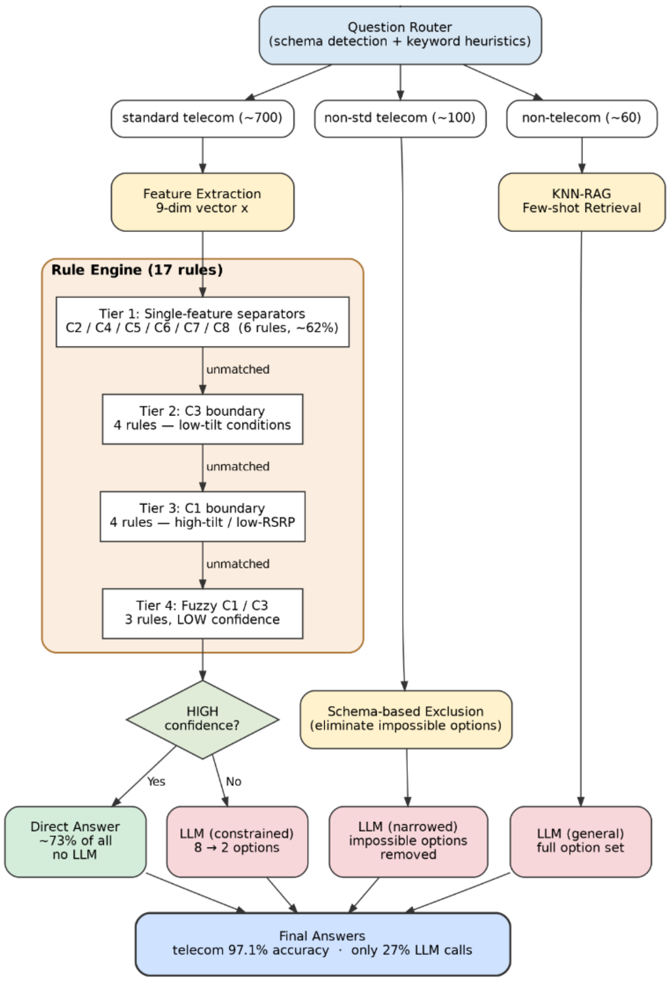

Fig. 2. Narrow-then-reason hybrid pipeline. All $\sim$700 standard telecom questions pass through the 17-rule engine. High-confidence rules resolve $\sim$73% of standard telecom questions without any LLM call; low-confidence residuals are sent to a constrained LLM with the option set narrowed from 8 to 2. Non-standard telecom and non-telecom questions are routed to general-purpose LLM inference.

(iv) atomic operators that are deterministic and numerical. Many telecom AIOps tasks fit this profile.

B. Capability Ceilings as a Design Primitive

Our small-model-RAG experiments crystallise a lesson worth stating plainly: *retrieval cannot lift a model above its reasoning ceiling*. Effective LLM deployment begins by mapping each candidate model's ceiling on the target operators, then routing only ceiling-fit subtasks to it. This is the inverse of the popular "RAG fixes everything" heuristic.

C. The Fine-Tuning Trap

Fine-tuning is a natural approach when training data is available. Domain-specific supervised fine-tuning (SFT) or LoRA adaptation [16] could plausibly push telecom-specific accuracy into the 80–90% range. However, the competition's mixed-domain question set exposes a fundamental trade-off that makes fine-tuning structurally unable to win.

The dual-objective constraint. Approximately 20% of the scored questions fall outside the standard 8-option telecom format: 82 are pure general-knowledge items (mathematics, history, logic puzzles) and the remainder are non-standard telecom questions with a different option schema that appears only in the test set. Fine-tuning on narrow 5G diagnosis data triggers *catastrophic forgetting* [2], [16]: the model's performance on these general-knowledge questions degrades, partially or fully offsetting the domain gains. This phenomenon is well-documented in the LLM era—Ouyang et al. [3] report that RLHF alignment incurs an "alignment tax" on certain benchmarks, and Luo et al. [4] show that continual fine-tuning leads to measurable forgetting on previously mastered tasks.

The competition leaderboard as evidence. The competition itself constitutes a large-scale natural experiment. Fine-tuning was a readily available strategy—indeed, our own codebase includes a complete LoRA pipeline [17]. While some teams achieved higher raw leaderboard scores, the competition's reproducibility review disqualified solutions that could not generalise to unseen data—confirming the structural prediction that fine-tuning trades non-telecom accuracy for telecom gains, and mixed-domain scoring penalises this trade-off.

Why the hybrid avoids this trap. Our design sidesteps the dilemma entirely. The rule engine handles the structured telecom questions—where fine-tuning would help—while the LLM remains un-fine-tuned and retains its full generalist capability for the non-telecom portion. The hybrid thus achieves *domain specialisation without generalist degradation*, a property that no monolithic fine-tuned model can guarantee.

D. Narrow-then-Reason as a General LLM Enhancement

Beyond telecom, the *narrow-then-reason* pattern constitutes a general method for improving LLM effectiveness on any task that admits partial deterministic structure. The mechanism is straightforward: rules shrink the hypothesis space from $|\mathcal{Y}|$ candidates to a subset $|\mathcal{Y}'| \ll |\mathcal{Y}|$; the LLM then reasons over $\mathcal{Y}'$ with a higher posterior concentration per candidate, yielding strictly higher expected accuracy than reasoning over $\mathcal{Y}$ directly. In our case, narrowing from 8 options to 2 raises the LLM's effective accuracy on residual samples substantially above its unconstrained baseline.

The applicability conditions are modest: (i) a portion of the label space can be ruled out by deterministic checks (thresholds, invariants, business rules), (ii) the label set is finite and semantically meaningful, and (iii) domain knowledge is available to encode as constraints. These conditions hold across many practical domains: medical triage (lab-value thresholds narrow differential diagnoses before an LLM reasons over the remainder), financial compliance (regulatory rules eliminate impossible categories before LLM review), and industrial quality control (sensor thresholds flag anomaly types before LLM root-cause analysis). In each case, rules do not *replace* the LLM—they make it better by focusing its reasoning capacity where it matters.

More broadly, rules and LLMs exhibit complementary failure modes: rules fail on ambiguous, context-dependent cases; LLMs fail on precise, deterministic operations. Combining them Pareto-dominates either component alone in accuracy, cost, latency, and auditability.

IX. Conclusion

Our 17-rule hybrid reached 97.1% telecom accuracy (93.85% overall), decisively outperforming frontier LLMs (<77%) and achieving the Track 1 championship among 200+ teams. The result is not a refutation of LLMs but a refinement of how to use them: respect capability ceilings, use rules to narrow before asking them to reason, and preserve generalist strengths by delegating domain-specific decisions to auditable symbolic components. The *narrow-then-reason* pattern generalises to any domain where partial deterministic structure coexists with residual ambiguity. Code is available at [17].

Acknowledgment

Rule synthesis was co-developed with Claude Opus 4.5 acting as a reasoning partner.

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